PREDICTIVE ANALYSIS OF FUTURE FRAUD RISK IN DIGITAL PAYMENT TRANSACTIONS

PROBLEM STATEMENT:

With the rapid growth of digital payment systems, the risk of fraudulent transactions has increased significantly, posing serious financial and security challenges. Traditional fraud detection methods are often rule-based and fail to identify new and evolving fraud patterns.

Therefore, there is a need to develop an intelligent system that can analyze large volumes of transaction data and accurately detect fraudulent activities in real time. This project aims to build a machine learning-based model that can effectively classify transactions as fraudulent or safe, improve detection accuracy, and minimize financial losses.

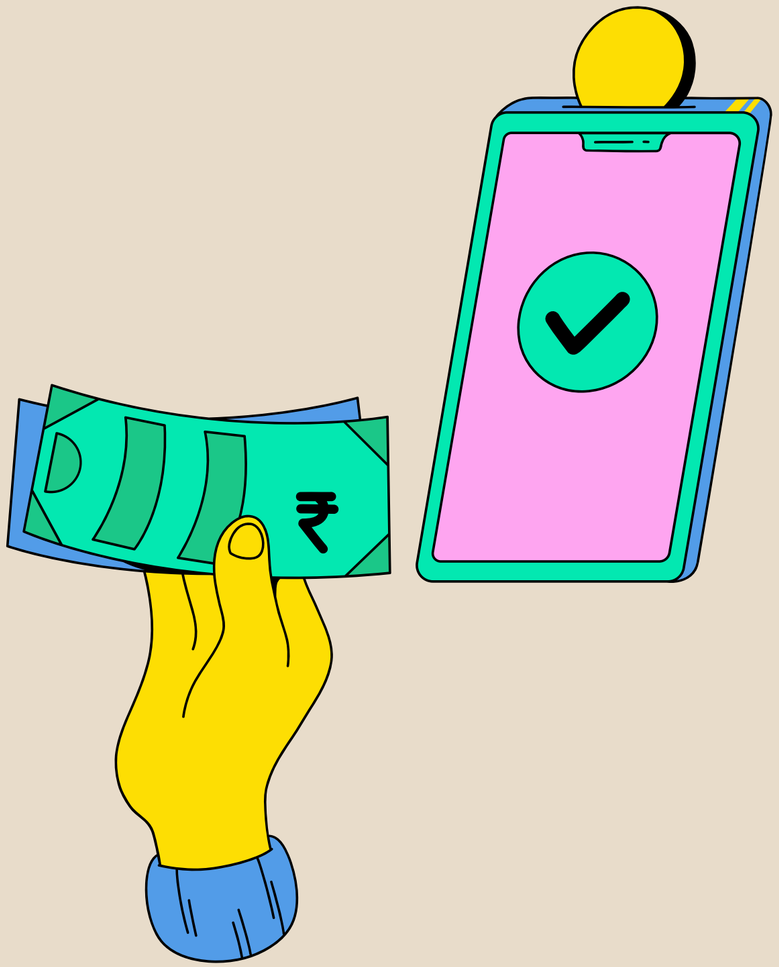
PROJECT EXPLANATION

Working of the System:

- The system takes transaction data as input (amount, balances, transaction type, etc.).
- Data is preprocessed by removing unnecessary columns and converting categorical values into numerical form.
- A new feature (balance_diff) is created to capture unusual balance changes.
- Since fraud data is very less, SMOTE technique is used to balance the dataset.
- The processed data is then used to train machine learning models.
- The trained models learn patterns of fraudulent and normal transactions.
- Finally, the system predicts whether a new transaction is:
  - Fraud (1)
  - Not Fraud (0)

Goal:

To create a reliable and accurate fraud detection system that can help reduce financial risks.



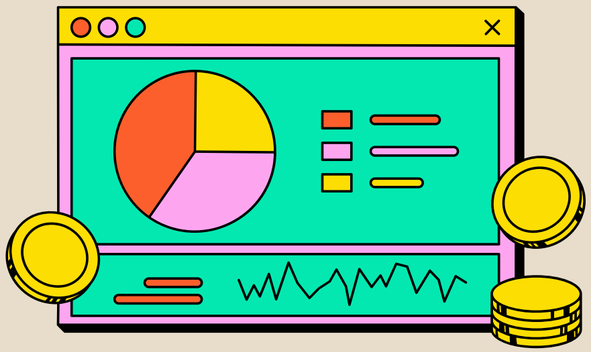
ALGORITHM(S) USED

LOGISTIC REGRESSION

- A basic classification algorithm used to predict binary outcomes.
- It calculates the probability of a transaction being fraud.
- Simple, fast, and easy to interpret.
- Used as a baseline model.

Role in Project:

- Used as a baseline model for comparison and provides basic prediction performance.

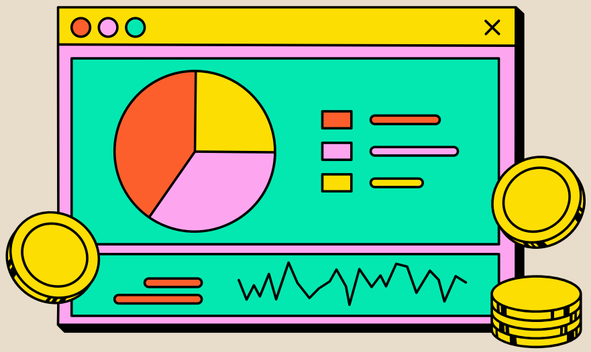


ALGORITHM(S) USED

RANDOM FOREST

- An ensemble learning algorithm that uses multiple decision trees.
- Each tree makes a prediction, and the final result is based on majority voting.
- Handles large datasets and complex patterns effectively.
- Reduces overfitting and improves accuracy.

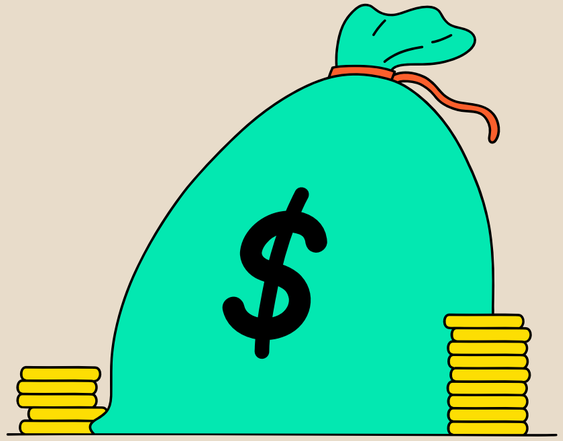
Role in Project:
This model gives the best performance and is used as the main prediction model.



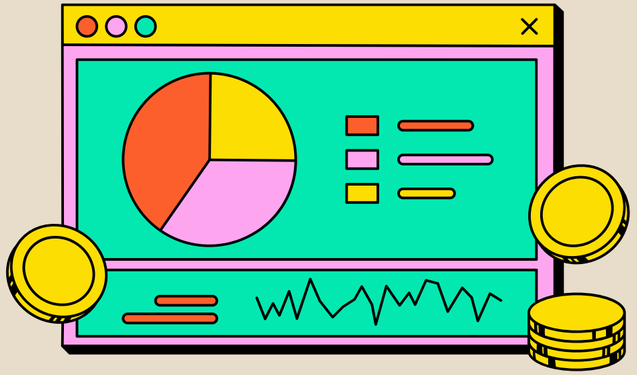
ALGORITHM(S) USED

GRADIENT BOOSTING

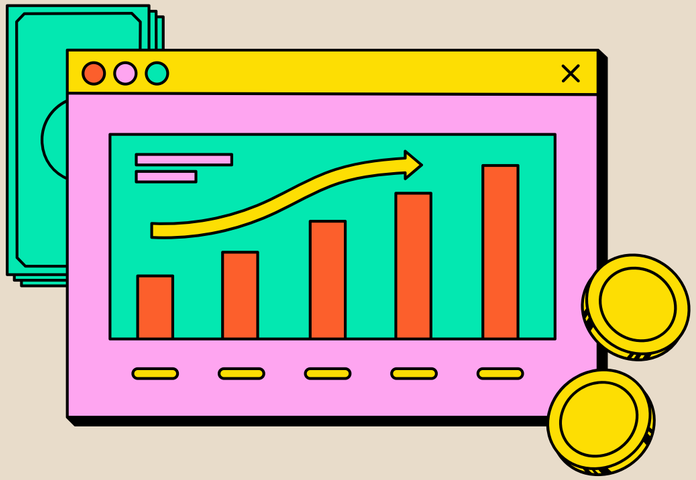
- Builds models sequentially, where each model corrects the errors of the previous one.
- Very powerful for complex datasets.
- Provides high prediction accuracy.



Role in Project:
Used to compare performance with Random Forest and improve results.



MODEL PERFORMANCE METRICS



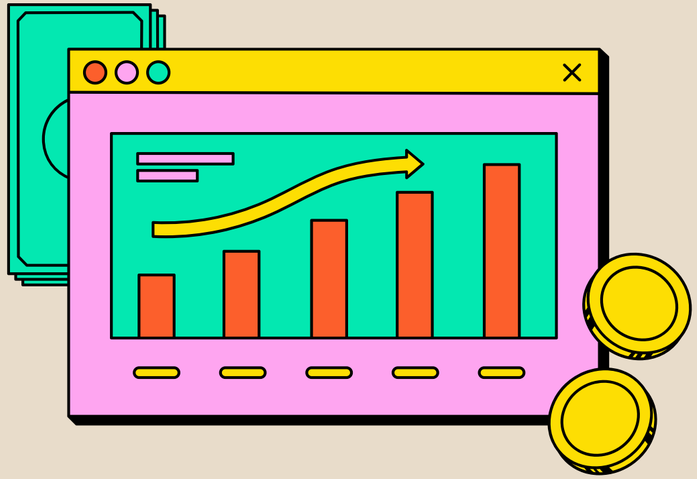
1) Accuracy:
Represents the overall percentage of transactions correctly classified as fraud or safe by the model.

2) Precision:
Indicates how accurately the model identifies fraud by measuring the proportion of predicted fraud cases that are truly fraudulent.

3) Recall:
Reflects the model's effectiveness in detecting actual fraud cases by measuring how many are correctly identified.

4) F1 Score:
Provides a balanced evaluation of the model by combining both precision and recall into a single performance metric.

MODEL PERFORMANCE (ACCURACY %)



Logistic Regression

- Accuracy: ~ 94% – 96%
- Basic performance, used for comparison

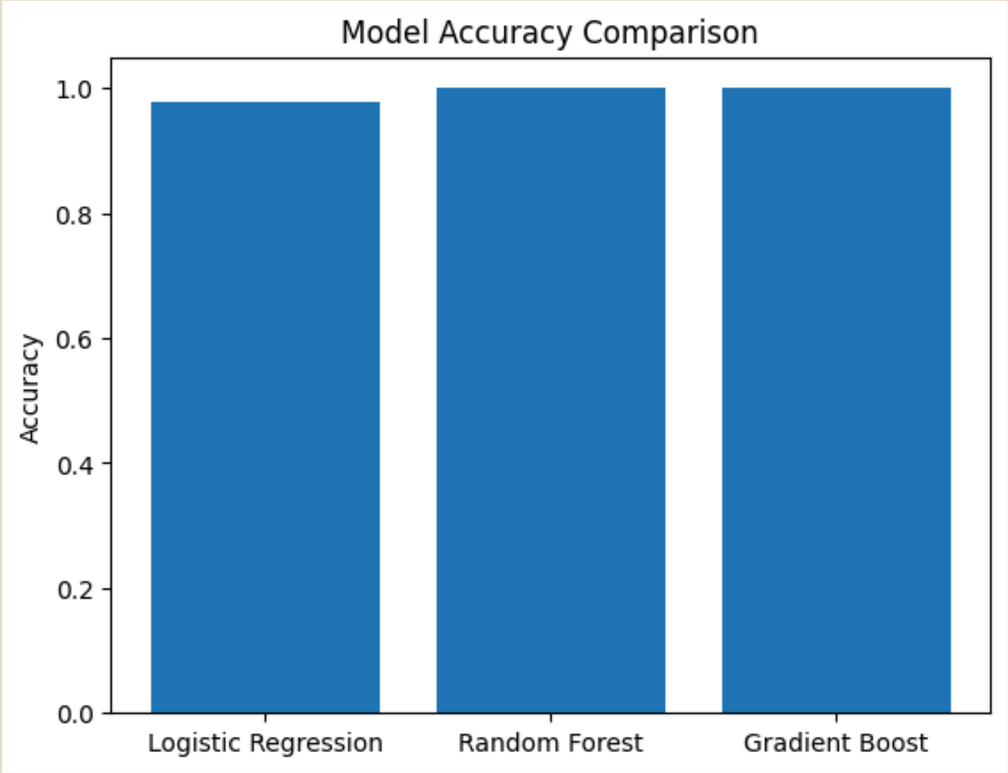
Random Forest

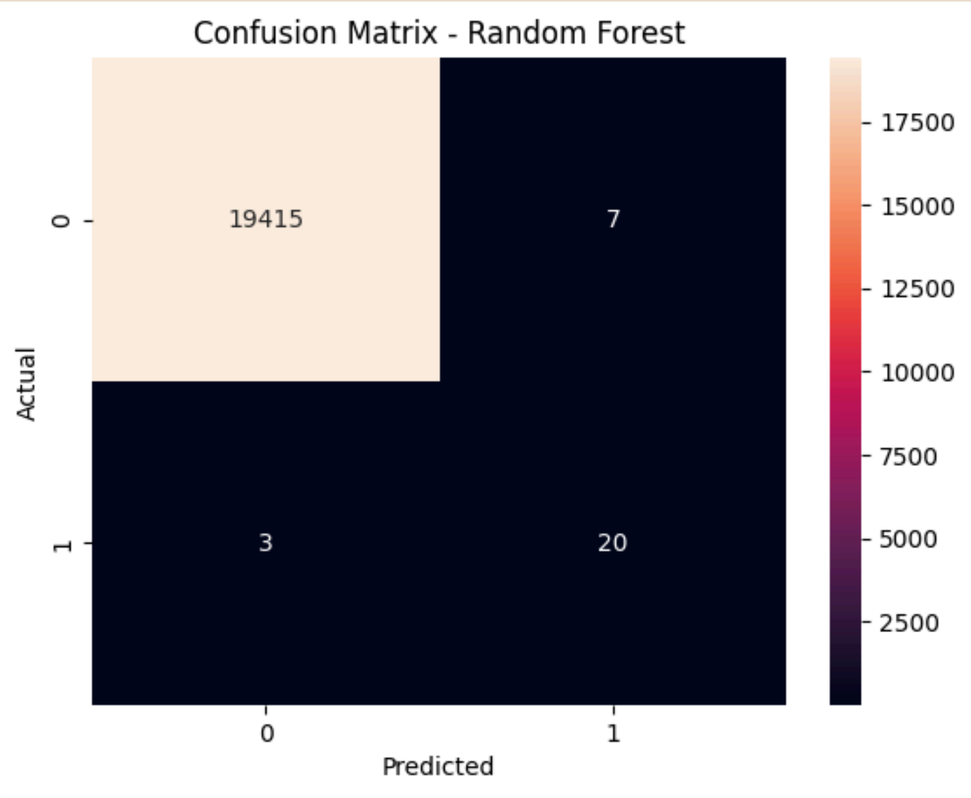
- Accuracy: ~ 98% – 99%
- Best performing model

Gradient Boosting

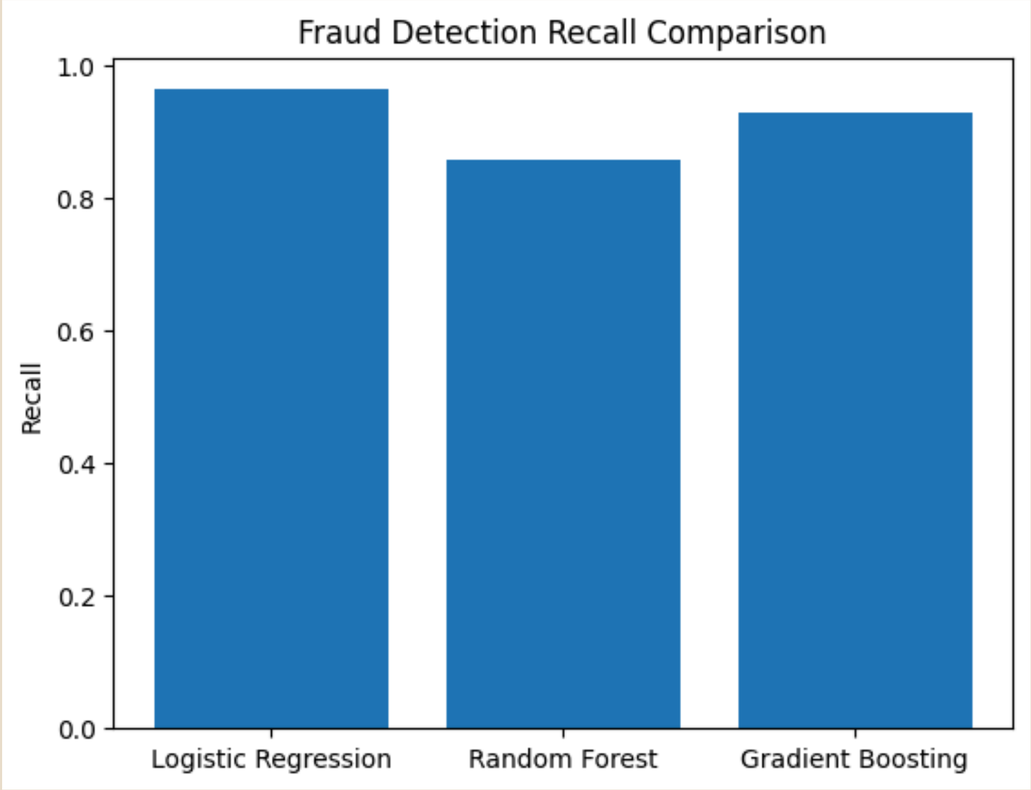
- Accuracy: ~ 97% – 98%
- High performance, slightly below Random Forest

OUTPUT:


Model Accuracy Comparison
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression
Random Forest
Gradient Boost


Confusion Matrix - Random Forest
Actual
0
1
19415
7
3
20
0
1
Predicted
17500
15000
12500
10000
7500
5000
2500

OUTPUT:


Fraud Detection Recall Comparison
Recall
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression
Random Forest
Gradient Boosting

PREDICTION:

```
📌 ENTER TRANSACTION DETAILS:

Enter step: 10
Enter amount: 90000
Enter oldbalanceOrg: 90000
Enter newbalanceOrig: 0
Enter oldbalanceDest: 0
Enter newbalanceDest: 90000
Enter type_CASH_OUT: 0
Enter type_DEBIT: 0
Enter type_PAYMENT: 0
Enter type_TRANSFER: 1
```

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Raw Probability: 0.6076234405293838

==================================================
🔍 FRAUD ANALYSIS RESULT
==================================================
Prediction: FRAUD 🚨
Fraud Risk Probability: 60.76%
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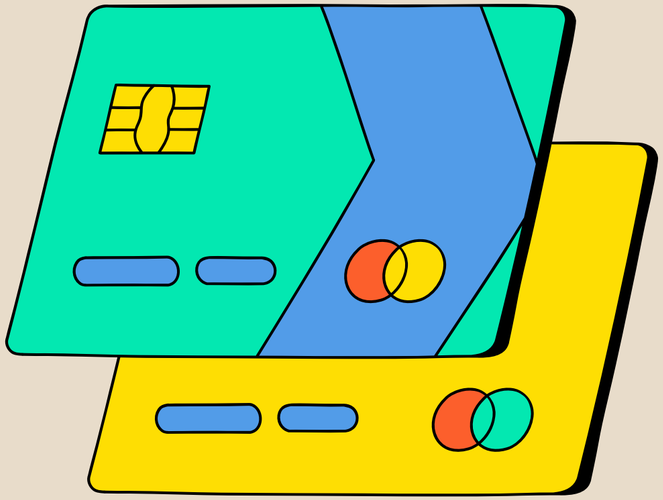

CONCLUSION

This project successfully demonstrates the use of machine learning techniques to detect fraudulent transactions in digital payment systems. Through effective data preprocessing, feature engineering, and handling of imbalanced data using SMOTE, the models were trained to accurately classify transactions as fraud or safe.

Among the implemented algorithms, Random Forest achieved the best performance, providing high accuracy and strong fraud detection capability. The use of evaluation metrics such as precision, recall, and F1 score ensured a reliable assessment of the model, with emphasis on recall to minimize undetected fraud cases.

Overall, the project highlights the potential of machine learning in enhancing financial security and reducing fraud-related risks, with scope for further improvements using advanced models and real-time systems.

THANK YOU

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